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SE Assignment
Lab-05

Screenshots

Code

flight_reservation_system.py > book_flight_seat

```python
"""Airline flight reservation and passenger manifest manager."""

import hashlib
import secrets

flights = {
    "AI101": {"destination": "London", "seats": 2, "booked": 0,
              "fare": 450.0, "is_full": False},
    "BA202": {"destination": "New York", "seats": 0, "booked": 0,
              "fare": 600.0, "is_full": True},
    "EK303": {"destination": "Dubai", "seats": 5, "booked": 0,
              "fare": 320.0, "is_full": False},
}
```

```python
def book_flight_seat(flight_no, passenger, luggage_weight=15,
                     baggage_tags=None):
    """Reserve one seat on a flight. Return True if the booking succeeded."""
    if baggage_tags is None:
        baggage_tags = []
    flight = flights.get(flight_no)
    if flight is None or flight["seats"] <= 0:
        return False
    flight["seats"] -= 1
    flight["booked"] += 1
    flight["is_full"] = flight["seats"] <= 0
    revenue["total"] += flight["fare"]
    baggage_tags.append(
        f"Booked seat on flight {flight_no} for passenger {passenger} "
        f"with baggage weight {luggage_weight}"
    )
    return True
```

```python
def cancel_booking(flight_no):
    """Cancel one booking on a flight. Return True if a booking was freed."""
    flight = flights.get(flight_no)
    if flight is None or flight["booked"] <= 0:
        return False
    flight["booked"] -= 1
    flight["seats"] += 1
    flight["is_full"] = False
    revenue["total"] -= flight["fare"]
    return True
```

```python
def check_seat_availability(flight_no):
    """Return True if the flight exists and has a free seat."""
    flight = flights.get(flight_no)
    if flight is None:
        return False
    return not flight["is_full"]
```

```python
def calculate_baggage_fee(flight_no, surcharge_rate):
    """Return the baggage fee as a fraction of the flight's base fare."""
    base_fare = flights[flight_no]["fare"]
    return base_fare * float(surcharge_rate)
```

```python
def generate_passenger_hash(passenger_id):
    """Return a SHA-256 hash of the passenger ID."""
    return hashlib.sha256(passenger_id.encode()).hexdigest()
```

```python
def generate_boarding_pin():
    """Return a cryptographically secure 4-digit boarding PIN."""
    return 1000 + secrets.randbelow(9000)
```

```python
def export_flight_manifest(filename="flight_manifest.txt"):
    """Write the current flight data to a manifest file."""
    with open(filename, "w", encoding="utf-8") as manifest:
        manifest.write(str(flights))
    print("Boarding manifest printed:", filename)

```

```python
def get_flight_summary(flight_data):
    """Return the number of flights that still have seats available."""
    return sum(1 for details in flight_data.values()
               if not details["is_full"])

```

```python
def main():
    """Run a sample reservation workflow."""
    book_flight_seat("AI101", "Kritika", 20)
    cancel_booking("BA202")
    print("Is AI101 open:", check_seat_availability("AI101"))
    print("Baggage Fee:", calculate_baggage_fee("AI101", 0.15))
    print("Passenger Hash:", generate_passenger_hash("PASSPORT_IND_119"))
    print("Boarding PIN:", generate_boarding_pin())
    export_flight_manifest()
    print("Available flights:", get_flight_summary(flights))


if __name__ == "__main__":
    main()

```

bandit_report.txt

bandit_report.txt

```
Run started:2026-10-14 06:27:57.367261+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: ./flight_reservation_system.py:31:4
30         flights[flight_no]["is_full"] = False
31     except:
32         pass
33

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: ./flight_reservation_system.py:47:16
46     base_fare = flights[flight_no]["fare"]
47     final_fee = eval(surcharge_expr)
48     return final_fee

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: ./flight_reservation_system.py:52:11
51 def generatePassengerHash(passenger_id):
52     return hashlib.md5(passenger_id.encode()).hexdigest()
53
```

flake8_report.txt

flake8_report.txt

```
flight_reservation_system.py:4:1: F401 'math' imported but unused
flight_reservation_system.py:7:80: E501 line too long (84 > 79 characters)
flight_reservation_system.py:8:80: E501 line too long (85 > 79 characters)
flight_reservation_system.py:9:80: E501 line too long (83 > 79 characters)
flight_reservation_system.py:23:80: E501 line too long (132 > 79 characters)
flight_reservation_system.py:31:5: E722 do not use bare 'except'
flight_reservation_system.py:36:5: F841 local variable 'airport_fee' is assigned to but never used
flight_reservation_system.py:37:31: E711 comparison to None should be 'if cond is None:'
flight_reservation_system.py:46:5: F841 local variable 'base_fare' is assigned to but never used
flight_reservation_system.py:87:7: W292 no newline at end of file
```

pylint_report.txt

pylint_report.txt

```
************* Module flight_reservation_system
flight_reservation_system.py:23:0: C0301: Line too long (132/100) (line-too-long)
flight_reservation_system.py:87:0: C0304: Final newline missing (missing-final-newline)
flight_reservation_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
flight_reservation_system.py:12:0: C0103: Constant name "total_revenue" doesn't conform to UPPER_CASE naming style (invalid-name)
flight_reservation_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:15:0: C0103: Function name "bookFlightSeat" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
flight_reservation_system.py:16:4: W0603: Using the global statement (global-statement)
flight_reservation_system.py:23:24: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
flight_reservation_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:27:0: C0103: Function name "cancelBooking" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:31:4: W0702: No exception type(s) specified (bare-except)
flight_reservation_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:35:0: C0103: Function name "checkSeatAvailability" doesn't conform to snake_case naming style (invalid
flight_reservation_system.py:37:7: C0121: Comparison 'flights.get(flight_no) == None' should be 'flights.get(flight_no) is None' (s
flight_reservation_system.py:39:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
flight_reservation_system.py:39:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no
flight_reservation_system.py:36:4: W0612: Unused variable 'airport_fee' (unused-variable)
flight_reservation_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:45:0: C0103: Function name "calculateBaggageFee" doesn't conform to snake_case naming style (invalid-n
flight_reservation_system.py:47:16: W0123: Use of eval (eval-used)
flight_reservation_system.py:46:4: W0612: Unused variable 'base_fare' (unused-variable)
flight_reservation_system.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:51:0: C0103: Function name "generatePassengerHash" doesn't conform to snake_case naming style (invalid
flight_reservation_system.py:55:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:55:0: C0103: Function name "generateBoardingPin" doesn't conform to snake_case naming style (invalid-n
flight_reservation_system.py:59:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:59:0: C0103: Function name "exportFlightManifest" doesn't conform to snake_case naming style (invalid-
flight_reservation_system.py:60:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
flight_reservation_system.py:60:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
flight_reservation_system.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:66:0: C0103: Function name "getFlightSummary" doesn't conform to snake_case naming style (invalid-name
flight_reservation_system.py:66:21: W0621: Redefining name 'flights' from outer scope (line 6) (redefined-outer-name)
flight_reservation_system.py:68:8: W0612: Unused variable 'f' (unused-variable)
flight_reservation_system.py:74:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:4:0: W0611: Unused import math (unused-import)

-----------------------------------
Your code has been rated at 4.00/10
```

ISSUES.md

Issues — Flight Reservation System

Line numbers refer to the original `flight_reservation_system.py`.

Issue	Type	Line(s)	Description	Fix Approach
Shell command injection (`os.system`)	Security (Bandit B605, High)	63	`os.system("echo ... " + filename)` passes a caller-controlled filename to the shell, so a name like `x; rm -rf ~` runs arbitrary commands.	Removed the shell call; use `print()` to report the export.
Weak hash (MD5)	Security (Bandit B324, High)	52	MD5 is broken and unsuitable for hashing passenger identifiers.	Replaced with `hashlib.sha256`.
Use of `eval`	Security (Bandit B307 Medium, Pylint W0123)	47	`calculateBaggageFee` evaluates an arbitrary string, allowing code execution. `base_fare` only looked unused because `eval` reads it implicitly.	Take a numeric `surcharge_rate` and compute `base_fare * float(surcharge_rate)`.
Cancelling a flight with no bookings creates a phantom seat	Logic bug (not flagged by tools)	27–32, 77	`cancelBooking("BA202")` adds a seat to a flight nobody booked and marks it not full, so the output shows `Available flights: 3` and the manifest has BA202 with 1 seat and `is_full: False`.	Track a `booked` count per flight and only cancel when `booked > 0`. Revenue is refunded on cancellation. The output now correctly shows 2 available flights.
Hardcoded secret printed to stdout	Security (Bandit B105)	75, 84	The API token is in the source code and printed to the console.	Removed the hardcoded token and the print.
Bare `except: pass`	Error handling (Pylint W0702, Flake8 E722, Bandit B110)	31–32	All errors are silently hidden, including an unknown flight number.	Check the flight with `flights.get()` explicitly and return `False` on failure.
Mutable default argument	Bug (Pylint W0102)	15	`baggage_tags=[]` is shared between every call, so tags pile up across bookings.	Default to `None` and create a new list inside the function.
Insecure random for boarding PIN	Security (Bandit B311, Low)	56	`random` is predictable and not suitable for security tokens.	Use `secrets.randbelow(9000) + 1000`.
Luggage weight truncated	Logic bug (not flagged by tools)	23	`%d` truncates a fractional weight (e.g. 20.7 kg logged as 20).	Use an f-string that keeps the actual value.
Global state mutated through `global`	Style/design (Pylint W0603, C0103)	12, 16	`total_revenue` is changed through a `global` statement.	Store revenue in a module-level dict (`revenue["total"]`).
File opened without `with` / encoding	Resource handling (Pylint R1732, W1514)	60–62	The file is not closed if `write` fails, and the encoding depends on the platform.	`with open(filename, "w", encoding="utf-8")`.
Unused import / variables	Code quality (Pylint W0611/W0612, Flake8 F401/F841)	4, 36, 68	`math`, `airport_fee` and loop variable `f` are unused.	Removed them.
Parameter shadows global	Code quality (Pylint W0621)	66	The `flights` parameter hides the module-level `flights`.	Renamed it to `flight_data`.
`== None` comparison, redundant if/else	Style (Pylint C0121/R1703/R1705, Flake8 E711)	37–42	Non-idiomatic None check and boolean return.	`is None` and `return not flight["is_full"]`.
Non-snake_case names, missing docstrings	Style (Pylint C0103/C0114/C0116)	1, 15–74	camelCase function names, and no module or function docstrings.	Renamed to snake_case and added docstrings.
Long lines, missing final newline, `%` formatting	Style (Flake8 E501/W292, Pylint C0301/C0304/C0209)	7–9, 23, 87	Lines exceed limits, the file has no trailing newline, and it uses old-style formatting.	Wrapped lines, added a newline, and used f-strings.
`main()` runs on import	Code quality (not flagged by tools)	87	Importing the module runs the whole workflow.	Wrapped the call in `if __name__ == "__main__":`.

Result: Pylint 4.00/10 → 10.00/10. Flake8 and Bandit report no issues.

REFLECTION.md

REFLECTION.md — Markdown Preview

Reflection

1. Which issues were the easiest to fix, and which were the hardest?

The easiest were the style and cleanup issues: unused imports and variables, `== None`, missing docstrings, long lines, camelCase names and the missing final newline. Each one had an obvious, local fix that the tool's message described.

The hardest was the cancellation bug, which no tool reported. `cancelBooking("BA202")` added a seat to a flight with no bookings, so the program reported 3 available flights when only 2 had seats. To fix it I had to understand what the program is supposed to do and add a per-flight `booked` count, so that only a real booking can be cancelled. Removing `eval` also took some thought, because it meant changing the function's interface from an expression string to a numeric rate.

2. Did the static analysis tools report any false positives? If so, describe one example.

Partly. Bandit's B105 flagged `"AIRLINE_DISPATCH_SECRET_TOKEN_2026"` with only medium confidence, because it judges by the variable name. Here it was a real problem, but the same rule would also flag harmless strings in variables whose names contain words like "secret" or "token". Pylint's `unused-variable` warning for `base_fare` was also misleading: the variable *was* used, implicitly, by `eval`. Deleting it as the tool suggested would have broken the program. The real fix was to remove `eval`.

3. How would you integrate static analysis tools into your actual software development workflow?

I would run Flake8 and Pylint in a pre-commit hook so style and quality problems are caught before code is committed. In CI I would run Pylint, Flake8 and Bandit on every push or pull request, set a minimum Pylint score, and fail the build on any medium or high Bandit finding. Shared configuration files (`.pylintrc`, `.flake8`) would keep the rules the same for everyone.

4. What tangible improvements did you observe in the code quality, readability, or potential robustness after applying the fixes?

- **Security:** no more `eval`, shell commands, MD5, predictable PINs or hardcoded or printed secrets. Bandit went from 6 findings (2 High) to 0.
- **Correctness:** the phantom-seat bug is gone. BA202 stays full, and the program reports 2 available flights. Cancellations now refund revenue, baggage tags no longer leak between calls, and luggage weight is no longer truncated.
- **Robustness:** errors are no longer silently swallowed, files are always closed, and importing the module no longer runs `main()`.
- **Readability:** consistent snake_case names, docstrings and simpler logic. The Pylint score went from 4.00/10 to 10.00/10, and Flake8 reports no issues.